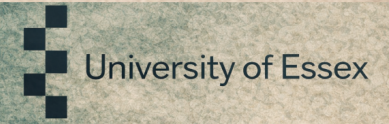


Effects of Short-Term Nature Exposure on Attention, Creativity, Heart Rate Variability, and Neural Oscillatory Activity



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Abstract

This study investigates whether **15 minutes** of exposure to **natural vs. urban** settings improves **attentional** control, **creativity**, heart rate variability (HRV), and neural oscillatory activity.

Using a **within-subjects, counterbalanced design** (N= 60, determined by G*Power *a priori* analysis), the research employs the Stroop task, Alternative Uses Task, Polar H7 sensors and Emotiv EPOC+ EEG.

The findings aim to clarify the integrated behavioural, physiological and neural mechanisms through which natural environments enhance cognitive functioning.



Nature vs Urban

Introduction

• **The Problem:** Sustained directed attention in daily life leads to cognitive impairment and mental fatigue (Kaplan 1995).

• **Theoretical Framework:** Attention Restoration Theory (ART): Nature restores attention via "**soft fascination**", bypassing executive function fatigue (Kaplan, 1995). Neurovisceral Integration Model: Autonomic, attentional and emotional systems are linked via a common neural self-regulation (Thayer & Lane, 2000).

• **Current Evidence:** Nature exposure is known to enhance cognition (Berman et al., 2008), creativity (Atchley et al., 2012), and parasympathetic activity (Park et al., 2010). From a cognitive neuropsychology perspective, focused attention is characterised by modulation of **alpha oscillations**: EEG and fMRI evidence that attentional filtering relies on cortical inhibitory process and prefrontal-parietal network activation (Klimesch, 1999; Corbetta & Shulman, 2002).

• **The Knowledge Gap:** Few studies simultaneously integrate behavioural, autonomic, and neural evidence, leaving the holistic mechanisms of nature-induced restoration underspecified.

• **Aim:** To examine whether 15 minutes of natural vs. urban exposure improves attentional control, creativity, HRV, and neural oscillatory activity in healthy adults.

Research Design & Method

• **Structure:** Within-subjects, counterbalanced design to mitigate order effects.

• **Sample:** N=60 adults (18-60 years), determined by **G*Power** *a priori* analysis.

• **Inclusion/Exclusion:** Normal vision and 15-min walking ability; no cardiovascular/neurological diseases or medications affecting autonomic function.

• **Measures:** Attentional Control = **Stroop Task** (interference scores) Creativity = **Alternative Uses Task** (fluency and originality) Autonomic Function = **Polar H7 (RMSSD)**; high-frequency HRV) Neural Activity = **Emotiv EPOC+ EEG (alpha power: 8-12 Hz)**

• **Procedure:** Two sessions (**≥48-hour washout**): Baseline (5-min HRV/EEG + tasks), 15-minute walk in a natural or urban setting, followed by post-walk measures.

• **Analysis:** Primary: **Repeated-measures ANOVA ($\alpha=.05$)** comparing conditions and time points. Exploratory: **Pearson correlations** between HRV changes and attentional performance.

Ethics

- **Approved** by University of Essex Online (Ref: Pending).
- Complies with **GDPR** and **BPS guidelines**.
- Data encrypted and pseudonymised.
- Participants will be **fully briefed** on EEG and heart rate monitoring equipment prior to consent.
- To mitigate equipment-induced stress, which can confound cortical activity measures, participants receive a technical briefing and pre-baseline **habituation period**.
- **Any discomfort** with equipment may be raised at any time and **apparatus removed**.

Limitations

- **Acute Effects:** 15-minute exposure may not generalise to long-term immersion.
- **Confounds:** Walking itself enhances divergent thinking (Oppezzo & Schwartz, 2014).
- **Mobile EEG:** Requires intensive ICA preprocessing to manage motion artefacts.
- **Environment:** Weather, noise, and air quality remain difficult to control in situ.

Conclusion

If supported, findings would suggest that even **brief, everyday nature exposure** produces **measurable autonomic and neural shifts** alongside **cognitive benefits** with direct implications for **workplace wellbeing, urban design, and mental health policy**.

References

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